

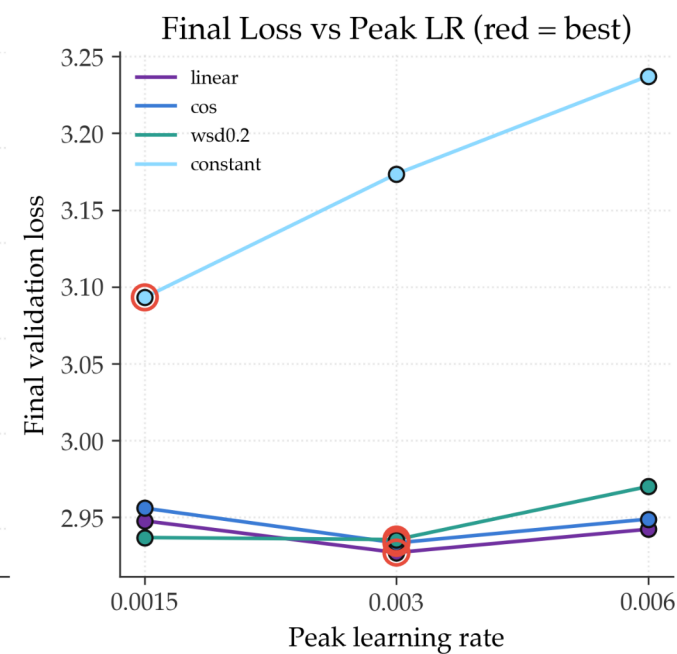
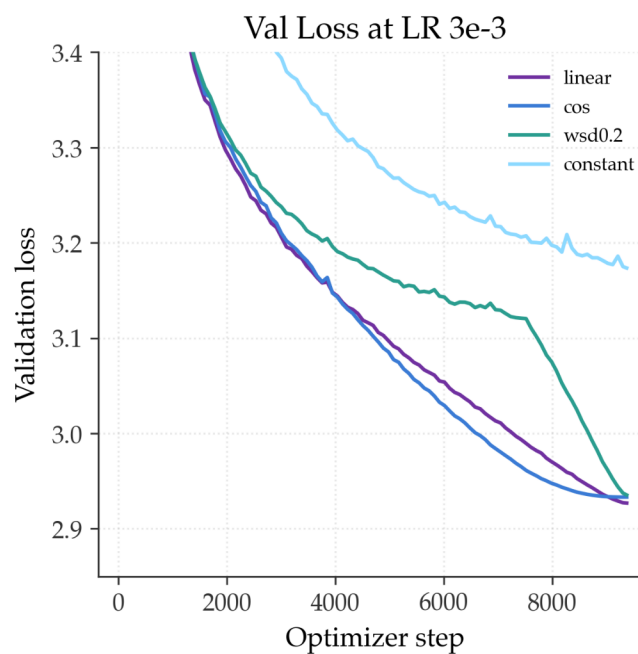
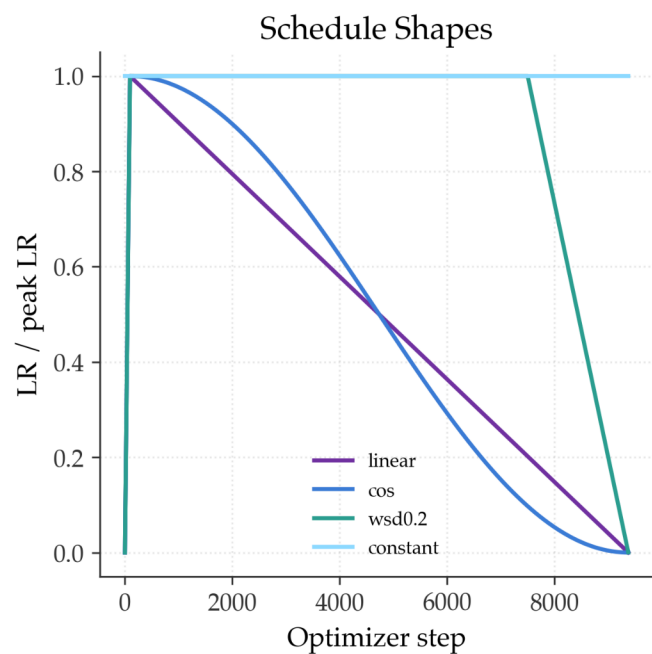
## Part (c): LR schedules and other optimizer hyperparameters (d8)

Compute-matched: d8, 614M tokens. 28 new runs; linear-schedule references reuse (a)/(b). Defaults unless noted: LR 3e-3, bs 64, warmup 1%, wd 0.1, AdamW (0.9, 0.95), clip 1.0, seeds 42. The 'constant' schedule ignores warmup.

| Experiment            | Result                                                                                                                     | Takeaway                                                                             |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Schedule x LR         | Decay matters most: constant (no warmup/decay) is +0.25 worse at LR 3e-3 and +0.17 worse even at its own best LR (1.5e-3). | Best LR 3e-3 for linear/cos; WSD shifts lower (1.5e-3 ~ 3e-3); constant wants lowest |
| WSD decay fraction    | 0.1 -> 2.956, 0.2 -> 2.935, 0.4 -> 2.920 (best overall at LR 3e-3), 1.0 (linear) -> 2.927                                  | Key extra hyperparameter; longer decay helps up to ~40%                              |
| Schedule x warmup     | All schedules gain from 4% warmup at LR 6e-3; ranking linear < cos < WSD stays                                             | Warmup x LR link from (b) holds for every schedule                                   |
| Schedule x batch size | Linear/cos prefer bs 32; WSD prefers 64 and degrades less at 128                                                           | Best batch size depends on schedule                                                  |
| beta2 / clipping      | beta2 0.99 best (2.922 @ 3e-3; 2.927 @ 6e-3); no clipping worst, esp. at LR 6e-3                                           | beta2 0.99 makes high LR safer; clipping helps at high LR                            |
| bs256 + warmup        | LR 6e-3: warmup 1% -> 3.083, 4% -> 2.962 (beats LR 3e-3 at 3.013)                                                          | bs256's high-LR failure in (b) was mostly too little warmup                          |
| Seed noise            | Default recipe, seeds 42/1/2: 2.927, 2.924, 2.930 (std 0.003, range 0.006)                                                 | Differences < ~0.006 are within noise                                                |

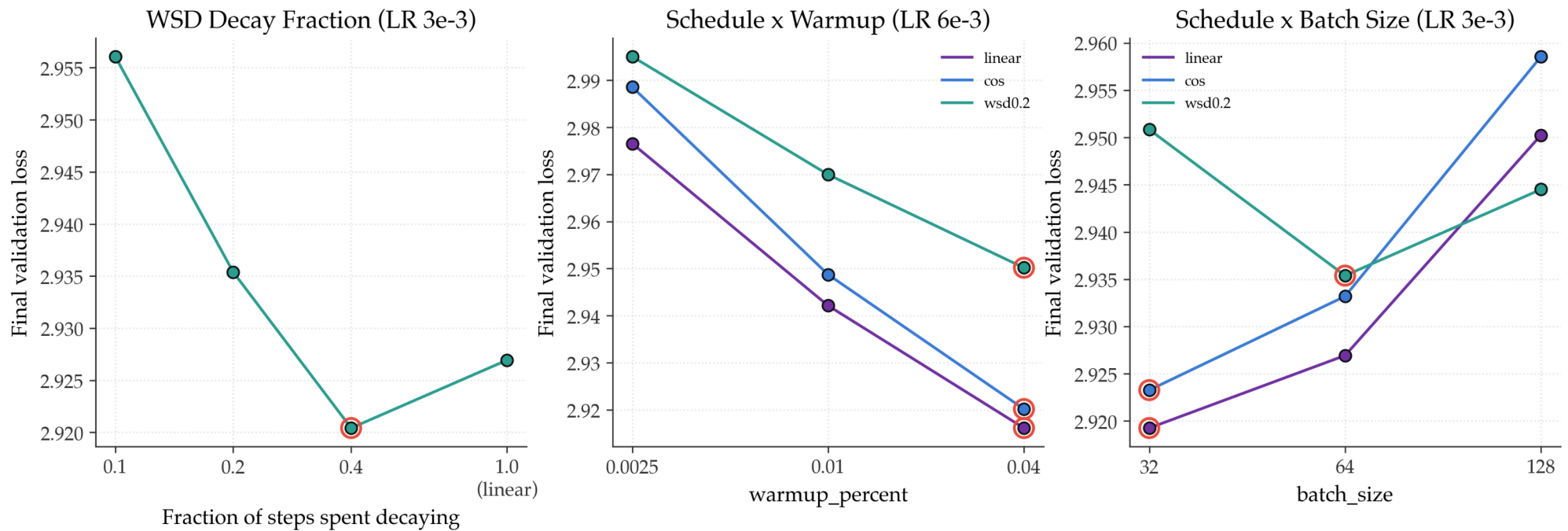
**Key additional hyperparameters: WSD decay fraction and Adam beta2. Tune together: LR with warmup and schedule.**

# Schedule x Learning rate



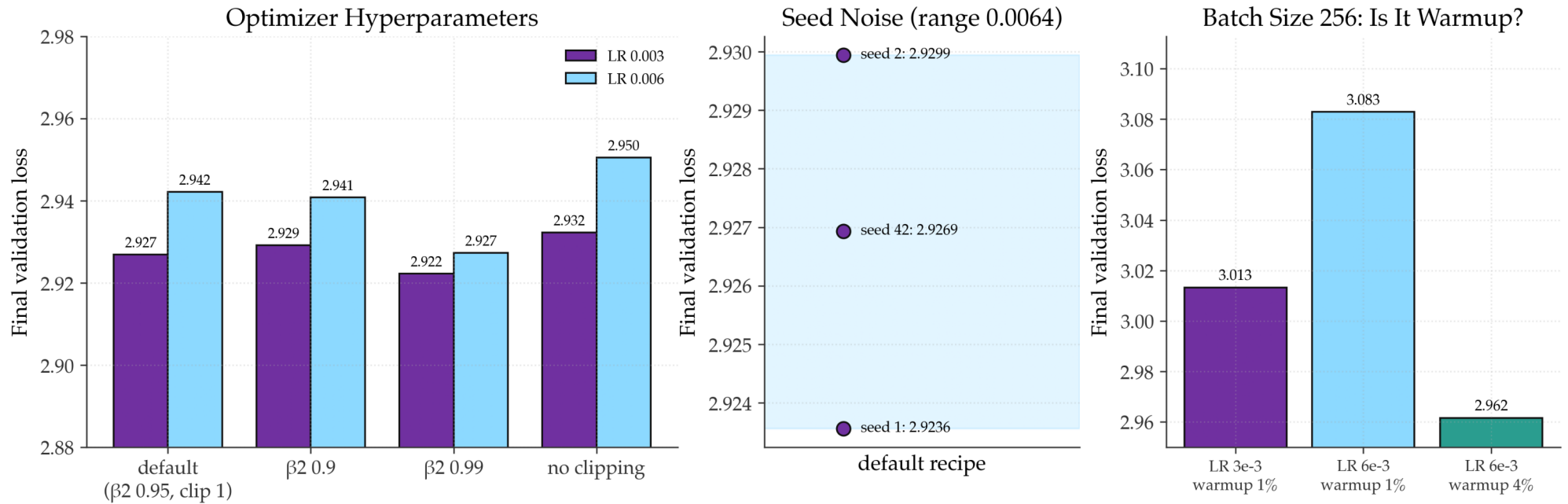
Left: LR multiplier over training (constant has no warmup). Middle: WSD's loss drops sharply during its final decay. Right: linear and cosine are best at LR 3e-3; WSD is flat from 1.5e-3 to 3e-3 and hurt more at 6e-3; constant is far worse and wants the lowest LR.

# WSD decay, warmup, and batch size



Left: WSD decay fraction is a key extra hyperparameter; 40% decay beats linear (2.920 vs 2.927, within seed noise).  
Middle: longer warmup helps every schedule at LR 6e-3. Right: WSD prefers bs 64 while linear/cosine prefer bs 32.

# Optimizer, seed noise, and batch size 256



Left:  $\beta_2$  0.99 is best and nearly removes the LR 6e-3 penalty; removing clipping hurts most at high LR. Middle: seed noise range 0.006. Right: with 4% warmup, bs256 at LR 6e-3 improves 3.083  $\rightarrow$  2.962, so large batches do tolerate higher LR once warmup is long enough.